```
clear
clc
close all

Mb = 1.5e6;              % Building mass (kg)
Kb = 15e6;               % Building stiffness (N/m)
zetab = 0.005;           % Building damping ratio
g = 9.81;                % Gravity (m/s^2)

Cb = 2*zetab*sqrt(Mb*Kb);

wn = sqrt(Kb/Mb);

fprintf('Natural frequency = %.4f rad/s\n',wn)
fprintf('Natural frequency = %.4f Hz\n',wn/(2*pi))
fprintf('Building damping = %.2f N*s/m\n\n',Cb)

om = linspace(0.1,6,5000);

% 1)
Gnum = ones(size(om));

Gden = Kb - Mb*om.^2 + 1i*Cb*om;

XF = abs(Gnum./Gden);

[peakResponse,peakIndex] = max(XF);

wr = om(peakIndex);

fprintf('Theoretical natural frequency = %.4f rad/s\n',wn)
fprintf('Resonant peak frequency       = %.4f rad/s\n',wr)
fprintf('Peak response                 = %.4e m/N\n\n',peakResponse)

figure(1)

semilogy(om,XF,'LineWidth',2)

hold on

xline(wn,'--','\omega_n','LineWidth',1.5)

plot(wr,peakResponse,'o','MarkerSize',8,'LineWidth',2)

title('Frequency Response of Building',...
      'FontSize',24,...
      'FontName','Times New Roman')

xlabel('Frequency (rad/s)',...
       'FontSize',24,...
       'FontName','Times New Roman')
```

```
ylabel('|X/F| (m/N)',...
       'FontSize',24,...
       'FontName','Times New Roman')

grid on

set(gca,'FontSize',18,...
        'LineWidth',2,...
        'FontName','Times New Roman')

set(gcf,'Position',[50 50 1000 600])

% 2)
ma = 0.01*Mb;             % Absorber mass
ca = 0;                   % No absorber damping

L = g/wn^2;

fprintf('Absorber mass = %.0f kg\n',ma)
fprintf('Tuned pendulum length = %.4f m\n\n',L)

% frequency response numerator

Gnum = ma*g*L ...
       - ma*L^2*om.^2 ...
       + 1i*ca*om;

% frequency response denominator

Gden = (Kb - (Mb+ma)*om.^2 + 1i*Cb*om) .* ...
       (ma*g*L - ma*L^2*om.^2 + 1i*ca*om) ...
       - (ma*L*om.^2).^2;

XF = abs(Gnum./Gden);

figure(2)

semilogy(om,XF,'LineWidth',2)

xline(wn,'--')

title('Building With Tuned Pendulum Absorber',...
      'FontSize',24,...
      'FontName','Times New Roman')

xlabel('Frequency (rad/s)',...
       'FontSize',24,...
       'FontName','Times New Roman')

ylabel('|X/F| (m/N)',...
       'FontSize',24,...
       'FontName','Times New Roman')

grid on
```

```
h = gca;

set(h,'FontSize',18,...
      'LineWidth',2,...
      'FontName','Times New Roman')

set(gcf,'Position',[50 50 1000 600])

% 3)
zetaValues = [0.005 0.01 0.02 0.05];

figure(3)
hold on

for j = 1:length(zetaValues)

    CbTest = 2*zetaValues(j)*sqrt(Mb*Kb);

    % Frequency response numerator

    Gnum = ma*g*L ...
           - ma*L^2*om.^2;

    % Frequency response denominator

    Gden = (Kb - (Mb+ma)*om.^2 + 1i*CbTest*om) .* ...
           (ma*g*L - ma*L^2*om.^2) ...
           - (ma*L*om.^2).^2;

    XF = abs(Gnum./Gden);


    semilogy(om,XF,'LineWidth',2)

end

xline(wn,'--')

title('Effect of Building Damping',...
      'FontSize',24,...
      'FontName','Times New Roman')

xlabel('Frequency (rad/s)',...
       'FontSize',24,...
       'FontName','Times New Roman')

ylabel('|X/F| (m/N)',...
       'FontSize',24,...
       'FontName','Times New Roman')

legend('\zeta_b = 0.005',...
       '\zeta_b = 0.01',...
       '\zeta_b = 0.02',...
```

```
       '\zeta_b = 0.05')

grid on

h = gca;

set(h,'FontSize',18,...
      'LineWidth',2,...
      'FontName','Times New Roman')

set(gcf,'Position',[50 50 1000 600])

% 4)
zetab = 0.005;

Cb = 2*zetab*sqrt(Mb*Kb);

massRatios = [0.01 0.03 0.05 0.075 0.10];

figure(4)
hold on

for j = 1:length(massRatios)

    % Absorber mass
    maTest = massRatios(j)*Mb;

    % Frequency response numerator

    Gnum = maTest*g*L ...
           - maTest*L^2*om.^2;

    % Frequency response denominator

    Gden = (Kb - (Mb+maTest)*om.^2 + 1i*Cb*om) .* ...
           (maTest*g*L - maTest*L^2*om.^2) ...
           - (maTest*L*om.^2).^2;

    XF = abs(Gnum./Gden);

    semilogy(om,XF,'LineWidth',2)

end

xline(wn,'--')

title('Effect of Absorber Mass',...
      'FontSize',24,...
      'FontName','Times New Roman')

xlabel('Frequency (rad/s)',...
       'FontSize',24,...
       'FontName','Times New Roman')
```

```
ylabel('|X/F| (m/N)',...
       'FontSize',24,...
       'FontName','Times New Roman')

legend('m_a = 0.01M_b',...
       'm_a = 0.03M_b',...
       'm_a = 0.05M_b',...
       'm_a = 0.075M_b',...
       'm_a = 0.10M_b')

grid on

h = gca;

set(h,'FontSize',18,...
      'LineWidth',2,...
      'FontName','Times New Roman')

set(gcf,'Position',[50 50 1000 600])


%  5)
zetab = 0.005;

Cb = 2*zetab*sqrt(Mb*Kb);

ma = 0.01*Mb;

% Absorber damping values
caValues = 500:500:9500;

% Pendulum length values
LValues = linspace(0.951*L,...
                   1.049*L,...
                   30);

% Frequency range around resonance
omTest = linspace(2.5,3.8,2000);

% initialize values
bestResponse = inf;

bestCa = 0;

bestL = 0;


% damping and length varying

for j = 1:length(caValues)

    caTest = caValues(j);
```

```
    for k = 1:length(LValues)

        LTest = LValues(k);

        % Frequency response numerator

        Gnum = ma*g*LTest ...
               - ma*LTest^2*omTest.^2 ...
               + 1i*caTest*omTest;

        % Frequency response denominator

        Gden = (Kb - (Mb+ma)*omTest.^2 + 1i*Cb*omTest) .* ...
               (ma*g*LTest - ma*LTest^2*omTest.^2 ...
               + 1i*caTest*omTest) ...
               - (ma*LTest*omTest.^2).^2;

        XF = abs(Gnum./Gden);

        % Maximum response in frequency range
        maxResponse = max(XF);

        if maxResponse < bestResponse

            bestResponse = maxResponse;

            bestCa = caTest; % current ca value being tested

            bestL = LTest; % current L value being tested

        end

    end

end
```

```
fprintf('Part 5 Results:\n')
fprintf('Best absorber damping = %.0f N*m*s/rad\n',bestCa)
fprintf('Best pendulum length = %.4f m\n',bestL)
fprintf('Original pendulum length = %.4f m\n',L)
fprintf('Maximum response = %.4e m/N\n\n',bestResponse)

%  Plot Graphs

% Frequency response numerator
```

```
Gnum = ma*g*bestL ...
       - ma*bestL^2*om.^2 ...
       + 1i*bestCa*om;

% Frequency response denominator

Gden = (Kb - (Mb+ma)*om.^2 + 1i*Cb*om) .* ...
       (ma*g*bestL - ma*bestL^2*om.^2 ...
       + 1i*bestCa*om) ...
       - (ma*bestL*om.^2).^2;

XF = abs(Gnum./Gden);

figure(5)

semilogy(om,XF,'LineWidth',2)

xline(wn,'--')

title('"Good" Broadband Response',...
      'FontSize',24,...
      'FontName','Times New Roman')

xlabel('Frequency (rad/s)',...
       'FontSize',24,...
       'FontName','Times New Roman')

ylabel('|X/F| (m/N)',...
       'FontSize',24,...
       'FontName','Times New Roman')

grid on

h = gca;

set(h,'FontSize',18,...
      'LineWidth',2,...
      'FontName','Times New Roman')

set(gcf,'Position',[50 50 1000 600])
```

```
%{
Part 3 Graph Discussion:  It can be seen in the graph that the amplitude of
the response curves for
 the building vary when you change the damping ratio. This makes
 the graph more smoothed out at the higher damping ratio such as 0.05.
Additionally, it is clear that each response curve approach the same notch
 value on the x-axis
%}

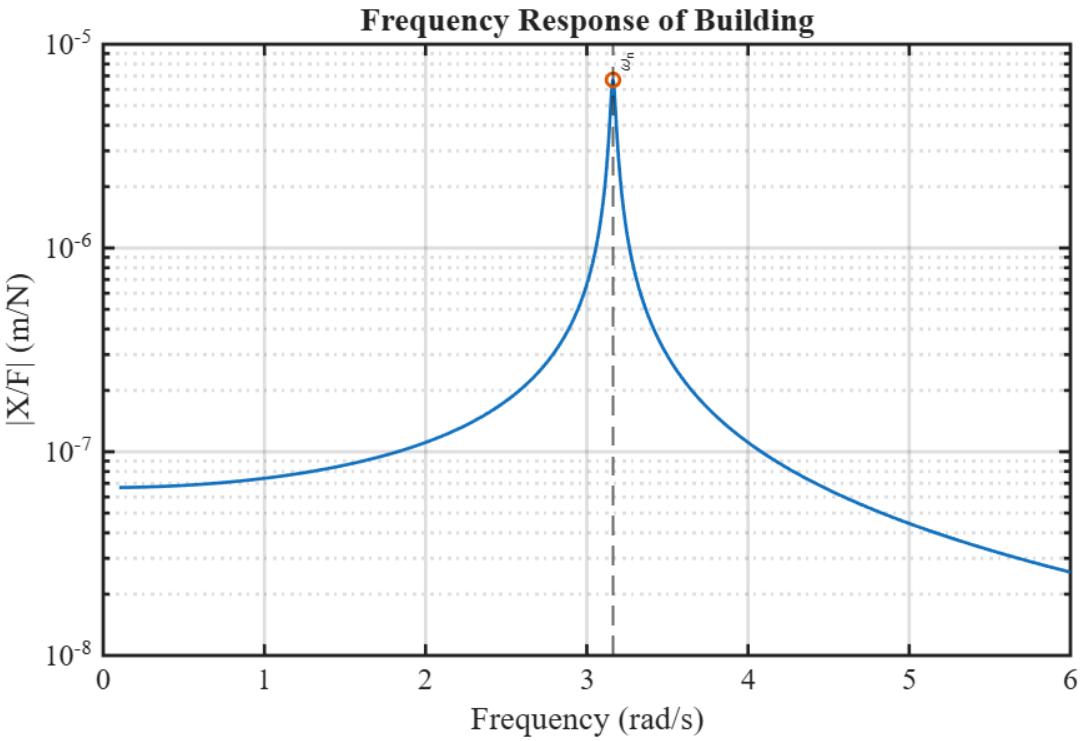
%{
Part 4 Graph Discussion: From the graph, the curves are slightly shifted
based on the absorber's mass. Additionally, it is shown that the amplitude
has a dependent relationship on the absorber mass because increasing
absorber mass equates to a higher amplitude of the response curve.
%}

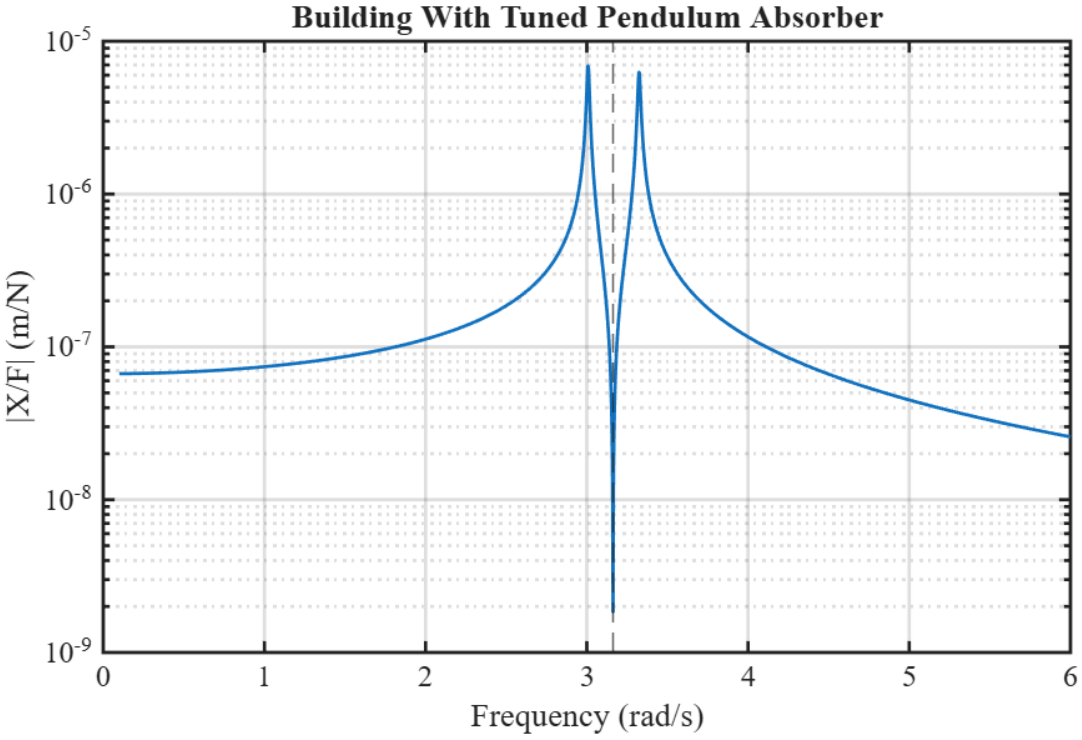
Natural frequency = 3.1623 rad/s
Natural frequency = 0.5033 Hz
Building damping = 47434.16 N*s/m

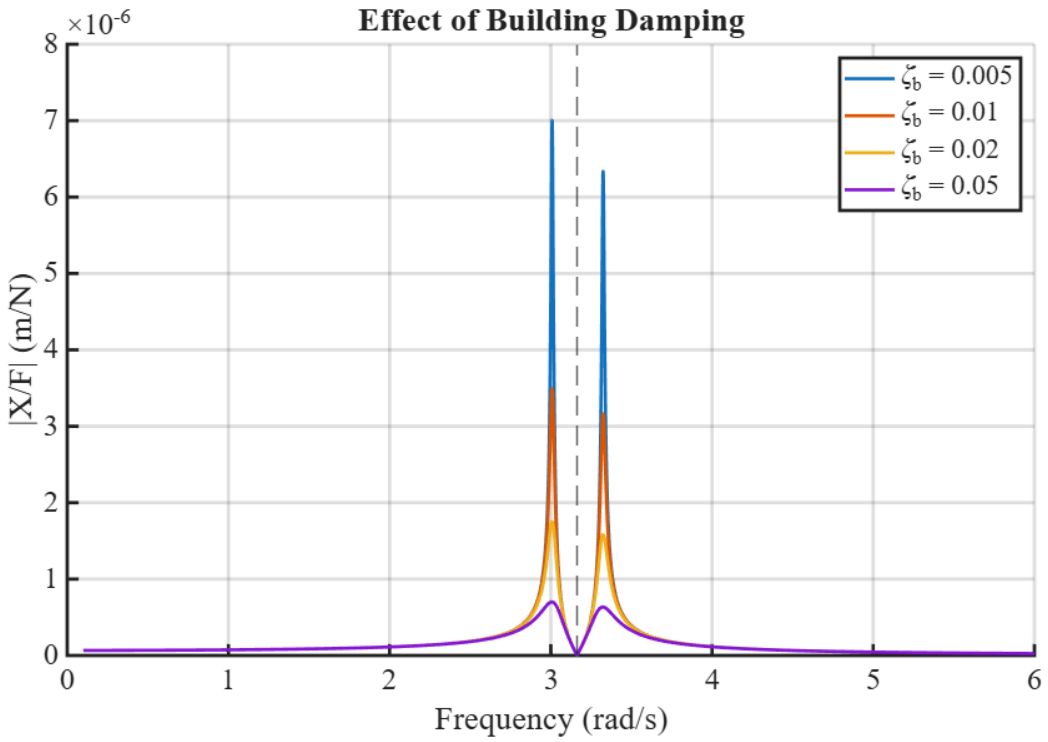
Theoretical natural frequency = 3.1623 rad/s
Resonant peak frequency       = 3.1627 rad/s
Peak response                 = 6.6632e-06 m/N

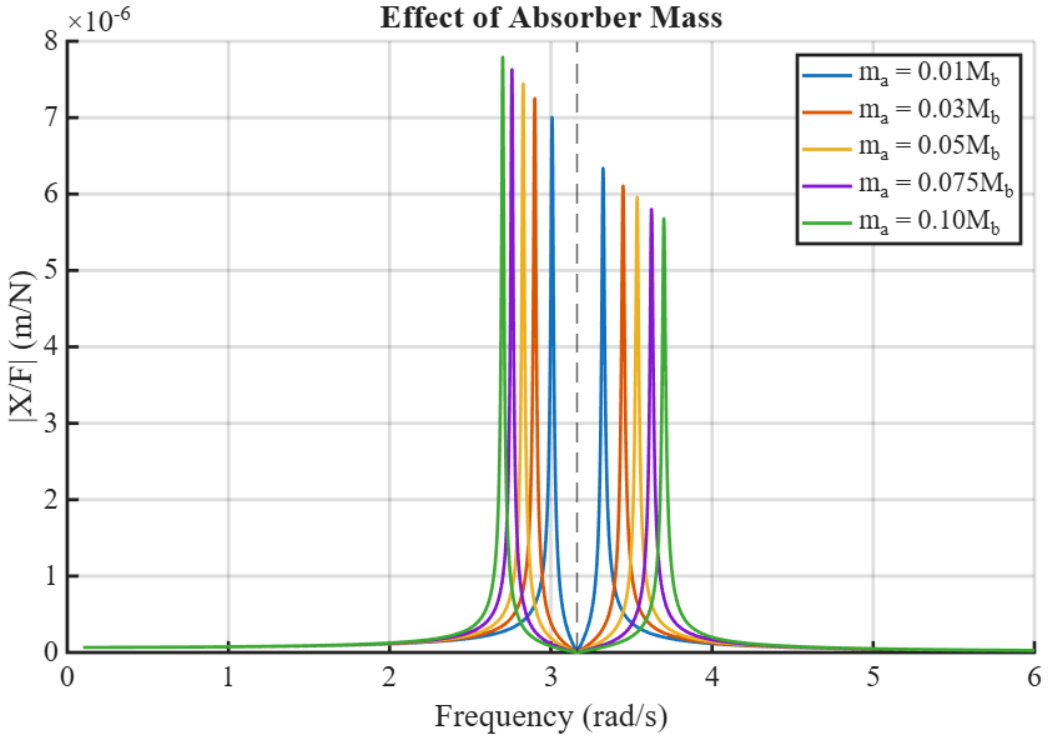
Absorber mass = 15000 kg
Tuned pendulum length = 0.9810 m

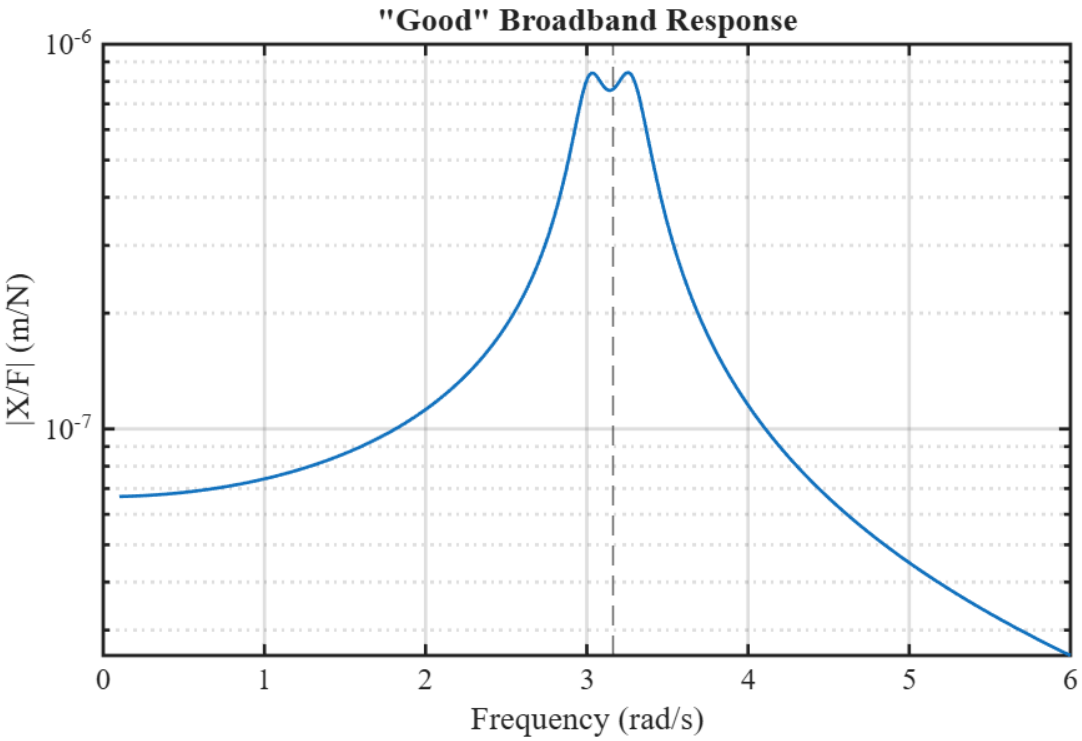
Part 5 Results:
Best absorber damping = 6000 N*m*s/rad
Best pendulum length = 1.0025 m
Original pendulum length = 0.9810 m
Maximum response = 8.4397e-07 m/N
```


Frequency Response of Building
ω_n
10^{-5}
10^{-6}
10^{-7}
10^{-8}
|X/F| (m/N)
0
1
2
3
4
5
6
Frequency (rad/s)


Building With Tuned Pendulum Absorber
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
|X/F| (m/N)
0
1
2
3
4
5
6
Frequency (rad/s)


Effect of Building Damping
×10-6
|X/F| (m/N)
Frequency (rad/s)
ζb = 0.005
ζb = 0.01
ζb = 0.02
ζb = 0.05


Effect of Absorber Mass
×10-6
|X/F| (m/N)
Frequency (rad/s)
ma = 0.01Mb
ma = 0.03Mb
ma = 0.05Mb
ma = 0.075Mb
ma = 0.10Mb



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